Multiarea Target Attention for Hyperspectral Image Classification

Huan Liu, Wei Li, *Senior Member, IEEE*, Xiang-Gen Xia, *Fellow, IEEE*, Mengmeng Zhang, and Ran Tao, *Senior Member, IEEE*

***Abstract*— In hyperspectral image (HSI) classification, objects corresponding to pixels of different classes exhibit varying size characteristics, which causes a challenge for effective pixelwise feature extraction and classification. In this article, we propose a novel multiscale model, called multiarea target attention (MATA). The proposed MATA uses an architecture that includes a shared feature extractor (FE) and classifier to capture multiscale spectral–spatial information effectively and efficiently. The FE uses a multiscale target attention module (MSTAM) to extract spectral–spatial information from target pixels and their similar pixels across multiscale areas, while L_2-normalization is used to address discrepancies between features of different scales. The classifier adopts a classwise decision weighting strategy to account for the varying sizes of different classes and the different contributions of semantic features at each scale to each class. Experimental results on five public HSI datasets demonstrate that the proposed MATA outperforms existing state-of-the-art single- and multiscale models, confirming its effectiveness and efficiency in HSI classification. Code is available at https://github.com/huanliu233/MATA.**

***Index Terms*— Hyperspectral image (HSI), multiscale feature extraction, target attention, Transformer.**

I. Introduction

With the advancement of deep learning, hyperspectral image (HSI) classification techniques have been advanced with various approaches [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11]. Many models use convolutional neural networks (CNNs) [12], [13], [14], [15], [16], [17]. Deep CNN for HSI (CNNHSI [18]) is the first to classify HSI using CNN, using only spectral information with three convolutional layers. Deep CNN with pixel-pair features (CNNPPFs [19]) indirectly uses spatial information by considering a pair of pixels, while CNN with randomized principle component analysis (RPCACNN [20]) and spatial–spectral feature extraction via CNN (SSCNN [21]) directly incorporate spatial information by performing randomized principal component analysis and traditional spectral–spatial extraction methods, respectively. Recently, the 3-D CNN framework (3DCNN [22]) is proposed to fully exploit the local spectral–spatial structure information. Furthermore, self-attention-based Transformer has been successfully applied in HSI classification [23], [24], [25], [26], [27], [28], [29]. Among these models, bidirectional encoder representation from transformers for HSI (BERTHSI [23]) is the first work to introduce self-attention to HSI classification, embedding a pixel sequence and feeding it into self-attention-based BERT modules to learn deep spectral–spatial features.

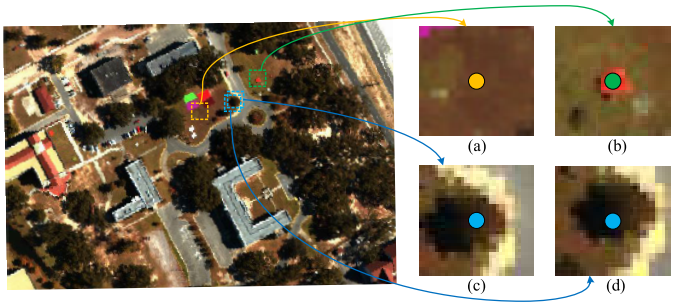


Fig. 1. (Left) Demonstration of limitation of single-scale image patch for HSI classification with MUFFL false-color image as an example. (a) and (b) Different classes have different sizes. (c) and (d) Different target pixels of the same class are in different areas of their corresponding patches.

The deep learning models mentioned above have made significant progress in hyperspectral image processing, but they face a challenge when dealing with target objects that vary in complexity, as depicted in Fig. 1. One limitation is the absence of consideration for size differences in target objects. These models typically use the same patch size to process objects of different sizes, which can make it difficult to identify and classify small target objects that are often overshadowed by the surrounding background. Another limitation is the lack of consideration for area differences within target objects. The area of a target object corresponding to a target pixel may be in a specific area of the HSI patch, and different target pixels may have different object areas. The pixels within a target object area serve to aid in the classification of target pixels, while pixels outside of the target area may contain interference information that could hinder accurate feature extraction.

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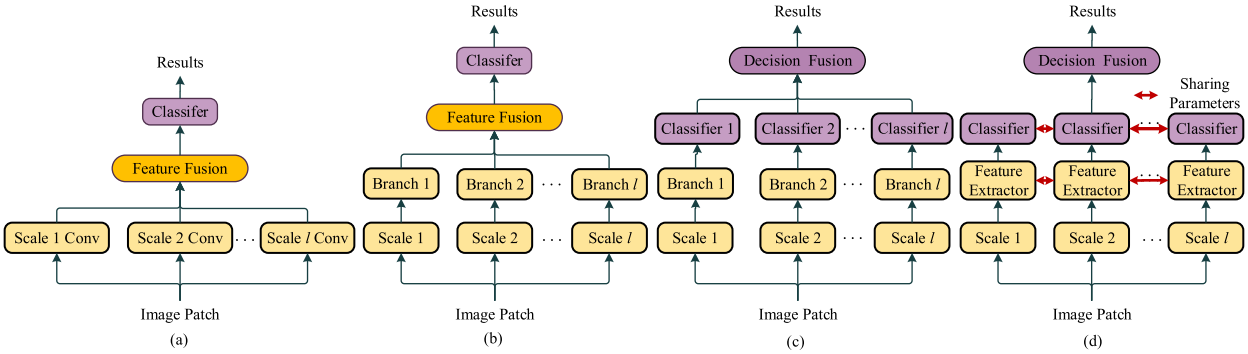


Fig. 2. Four categories of multiscale models for HSI classification. (a) Multiscale convolutional filter bank with feature fusion. (b) Multibranch FEs with feature fusion. (c) Multibranch FEs and classifiers with decision fusion. (d) One shared FE and classifier with decision fusion.

To overcome the limitations of single-scale deep learning models, researchers have proposed multiscale models that can process images at different scales. These models can capture the features of objects of different sizes and improve the model's robustness to accurately identify and classify target objects at different scales. One such model is the contextual deep CNN (CDCNN) [30], which performs spectral–spatial extraction on an HSI patch using a multiscale convolutional filter bank at the beginning and then uses more convolutional layers to further extract deep semantic information from the concatenated multiscale features. The multiscale CNNs (MS-CNNs [31]) use 1-D, 2-D, and 3-D convolutions and multiple kernels to extract deep multiscale spectral–spatial features with more representational ability. The diverse region-based CNN (DRCNN [17]) model uses different branches for different areas of the HSI patch within a target pixel to learn the spectral–spatial features of different areas and merge them to obtain the final features. In [32], multiscale spectral–spatial features with different spatial sizes in different layers are separately projected into decision space, and two models, side output decision fusion network (SODFN) and fully convolutional layer fusion network (FCLFN), are developed by fusing the probability vectors and logit vectors in decision layers, respectively. As a variant of CNN, graph convolutional network (GCN) can extract different types of spectral–spatial features. Based on that, FusCNNGCN [33] uses two branches including CNN and GCN to extract multiscale spectral–spatial features and fuse them by a full connection layer.

The existing multiscale models for HSI classification operate on the assumption that each scale provides valuable information for classification and can aid in target pixel identification through feature or decision fusion. However, it is possible that different scales or areas contribute varying amounts of discriminative information. In addition, most of the multiscale models use multiscale convolutional filter bank, or multibranch feature extractors (FEs) to extract multiscale features, which consumes abundant computational resources. To address these issues and thus extract multiscale spectral–spatial features more effectively and efficiently, this article proposes a novel multiscale model for HSI classification, called multiarea target attention (MATA) that uses a parameter-shared FE and classifier. The FE includes a multiscale target attention module (MSTAM) that can extract spectral–spatial information from a target pixel and its similar surrounding pixels. By feeding multiple parts of the HSI patch into the MSTAM, the model can extract multiscale spectral–spatial features, which are subsequently L_2-normalized to reduce discrepancies between different scales. The spectral–spatial features of each scale are then transformed into logit space using a parameter-shared classifier that consists of two linear projections. Finally, trainable weight vectors are used to fuse the multiscale logits of each class, and the final probability of each class is obtained through softmax activation.

This article has the following significant contributions.

1) The proposed MATA for HSI feature extraction and classification is the simplest multiscale architecture available, using only one parameter-shared FE and classifier, leading to parameter efficiency.
2) MSTAM is designed to extract spectral–spatial features of different areas to highlight the characteristics of a target object, taking into account that the object related to a target pixel may be part of an HSI patch.
3) Objects of different classes may have different sizes, and the extracted spectral–spatial features of areas of different sizes may contain different levels of reliable identification information. Therefore, trainable weight vectors are used in logits fusion to make more reliable class decision.

The remaining sections of this article are organized as follows. Section II presents the background and related works. Section III provides a detailed explanation of the proposed MATA. The experimental results are presented in Section IV. Finally, we conclude in Section V.

II. Background and Related Works

A. Taxonomy of Multiscale Models for HSI Classification

A taxonomy of multiscale models for HSI classification is proposed based on two points: 1) whether the multiscale features or decisions are fused and 2) for multiscale feature extraction, whether it uses multiscale convolutional filter bank, multiple feature extraction branches, or only one shared FE. These two points lead to four types as shown in Fig. 2.

Feature fusion is commonly used in many multiscale models. One effective way to extract and fuse multiscale features

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is to use a multiscale convolutional filter bank that involves convolutions with various kernel sizes. This approach is commonly used in several models, such as CDCNN [30], MS-CNNs [31], and pyramidal multiscale convolutional network (PMCN [34]), as shown in Fig. 2(a). These models perform multiscale feature extraction by applying multiple convolutions with different kernel sizes. Then, the resulting multiscale features are concatenated and passed through an FE to obtain the fused features. However, the effective information of a target object and interfering spatial information of other pixels in different classes may become entangled, which negatively impacts the accuracy of classification.

Many models adopt the strategy of using multiple branches to extract and fuse spectral–spatial features from multiscale patches, such as DRCNN [17], multifeature fusion network using CNN and GCN (MFGCN [35]), FusCNNGCN [33], and scale distillation network (SDNet [36]) [see Fig. 2(b)]. These feature extraction branches do not share any parameters and have different structures. To simultaneously use these multiscale spectral–spatial features, they concatenate and fuse these features before feeding them into the decision layer, which enhances the discriminant information. Using different branches enables separate feature extraction for different scales, allowing for targeted feature extraction and avoiding entanglement of multiscale features. However, using multiple branches increases the model complexity and imposes a heavy computational burden.

Unlike models that use feature fusion, some models use decision fusion, such as SODFN [32] and FCLFN [32] [as shown in Fig. 2(c)]. In the feature extraction process, these models use different branches to extract multiscale features, similar to the approach shown in Fig. 2(b). However, instead of fusing these multiscale features, they project them using linear classifiers into decision space separately and fuse the logits or probabilities in the decision layer. This approach allows for separate and complete extraction of discriminant information through different branches and classifiers, while completely avoiding the problem of information entanglement. However, this approach is still challenged by the number of branches that increases with the increase in scales, which imposes an even heavier computational burden than the type in Fig. 2(b) since it needs extra classifier branches.

The proposed MATA, which is the type shown in Fig. 2(d), uses the same FE for each scale or area and obtains multiscale decisions using the same linear projections. The decisions are then fused using classwise weighting. This approach results in a model with a fixed number of parameters, making it possible to feed the model with any number of multiscale or area inputs. Moreover, the model can learn different sizes of different classes and extract more robust spectral–spatial features through classwise weighting.

B. FE Schema

The core of multiscale models for HSI classification is FE. This component outputs spectral–spatial features from the complex HSI, which are then fed to the final classifier. The quality of the extracted spectral–spatial features is crucial in determining whether a target pixel can be correctly classified.

To date, two primary types of FEs for HSI classification are: 1) CNN-based approaches, which extract multiscale features using multiple branches of convolutional modules or multiple convolutions with different kernel sizes, and 2) transformer-like approaches, which process inputs using self-attention-like modules. We have chosen to follow the Transformer-like approach and design a shared FE because it is more flexible than the CNN-based approach. The Transformer-like approach can handle inputs of any image size, allowing for multiscale feature extraction to share a single extractor.

C. Multiscale Information Interaction Schema

In multiscale models, the interaction of discriminant information from different scales or areas during the process of feature or decision fusion results in the output of more discriminant information, achieving a greater-than-the-sum-of-its-parts effect. We categorize multiscale information interaction methods into two types: 1) equally treated approaches (e.g., CDCNN [30], DRCNN [17]), where multiscale features or decisions are equally treated, concatenated and projected to the fused space, added, or pooled, and 2) weighting approaches (e.g., multiscale feature extraction model [MSFE [37]])], where the importance of each scale is learnable and multiscale features or decisions are weighted pooled using different trainable weights for different scales.

We have chosen to follow the weighting approach as it better suits our problem, where different classes have different sizes and each area contributes differently to the discriminant information. This approach allows us to learn the importance of each scale and effectively interact with the extracted multiscale information. While equally treated approaches can enlarge the discriminant information, they are suboptimal as they do not consider the importance differences between scales.

III. Methodology

The architecture of the proposed MATA is illustrated in Fig. 3, including area division, multiscale feature extraction, and classwise weighted decision fusion (WDF).

A. Area Division Strategy

In HSI classification, the spectral information of a target pixel contains abundant discriminant information, but relying solely on this information may not achieve a satisfactory classification result. To mitigate this issue, information from surrounding pixels is also used, since some of them may belong to the same class as the target pixel and jointly form a target object. For convenience, an HSI patch centered on the target pixel is typically used as the input.

However, the size of a target object varies from class to class, and the position of a target object varies from pixel to pixel and may be in any area of the HSI patch. As shown in Fig. 1, pixels A and B belong to different classes, and their corresponding target objects have different sizes: pixel A has a larger target object than pixel B, and using a larger patch improves the classification accuracy of pixel A, while a smaller patch improves the classification accuracy of pixel B. Pixels C and D are in the same class, but their target objects have

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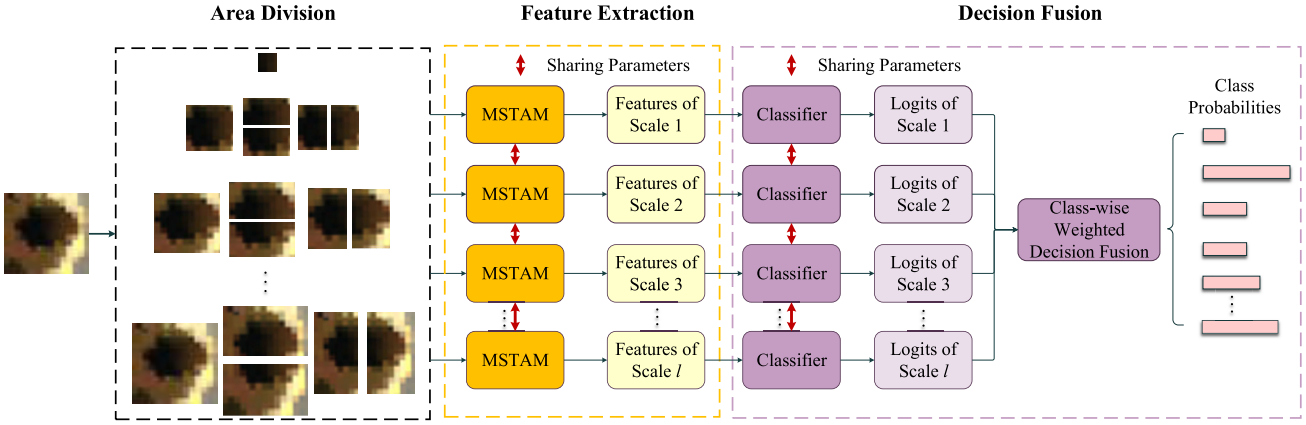


Fig. 3. Illustration of the proposed MATA for HSI classification, including area division strategy, feature extraction by one MSTAM, and classwise WDF.

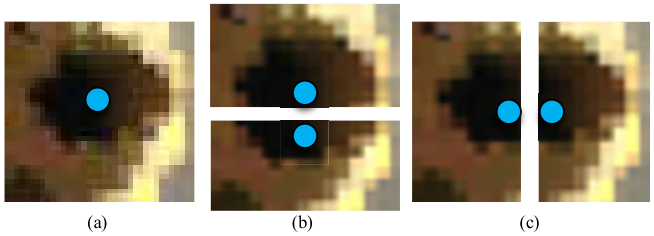


Fig. 4. Area division strategies for hyperspectral image classification. (a) Image patch. (b) Up and down areas. (c) Left and right areas.

different effective areas in their HSI patches: the corresponding object of pixel C is in the upper part of its image patch, while that of pixel D is in the left part of its image patch. To identify the areas of these objects, it is necessary to take into account the discriminant information from each area of the image patch.

To identify objects of different sizes, we use multiple HSI patches of different sizes centered on the target pixel. Specifically, patches with sizes $r \times r$, $1 \le r \le R$, are used as inputs to our models, where R is a hyperparameter that can be tuned. Moreover, to identify objects in different areas, patches of different sizes are divided into areas of different directions: up, down, left, and right parts of the patches, as shown in Fig. 4. For a patch of size $r \times r$, its corresponding patches for each direction are: up and down with size $(r+1)/2 \times r$, and left and right with size $r \times (r+1)/2$, respectively.

B. FE Design

Our FE is MSTAM, as shown in Fig. 5, which can process inputs with different image sizes. Once multiarea inputs corresponding to a target pixel are prepared, they are fed into MSTAM simultaneously.

1) Scaled Dot-Product Target Attention: Assuming that one of the multiarea inputs corresponding to a target pixel is $\mathbf{X}_i \in \mathbb{R}^{C_i \times m \times n}$, where m and n represent the height and width, respectively, and C_i is the number of channels in the input. During the calculation process, the 3-D tensor is converted into a pixelwise 2-D matrix $\mathbf{x}_i \in \mathbb{R}^{C_i \times mn}$. The value matrix $\mathbf{y} \in \mathbb{R}^{C_o \times mn}$ is then derived as follows:

$$\mathbf{y} = \mathbf{ReLU}(\mathbf{BN}(\mathbf{W}_o \mathbf{x}_i + \mathbf{b}_o)) \tag{1}$$

where $\mathbf{W}_o \in \mathbb{R}^{C_o \times C_i}$ and $\mathbf{b}_o \in \mathbb{R}^{C_o \times 1}$ are the learnable weights and biases, respectively, and C_o is the number of channels of the value matrix. **ReLU** and **BN** are the activation function (rectified linear unit) and normalization function (batch normalization), respectively. The purpose of using **ReLU** is to make the model nonlinear, while **BN** is used to normalize the samples to the same distribution to avoid internal covariate shift and speed up training convergence. These functions play important roles in deep learning and are commonly used in image processing models. This is different from the self-attention module as the latter does not use any normalization and activation functions in calculating the value matrix. Instead, it uses layer normalization and **GELU** in a feedforward network (FFN) after self-attention to generate nonlinear features. Layer normalization and **GELU** are successfully applied to language processing models, but it is difficult to train image processing models using them when plenty of training samples are not available. Since the number of training samples available for pixel-level HSI classification is generally limited, our models exclude the FFN in the attention module. Instead, we use **BN** and **ReLU** to compute the value matrix and find them empirically beneficial for better nonlinear feature extraction in pixel-level HSI classification.

The key matrix $\mathbf{z} \in \mathbb{R}^{C_o \times mn}$ is obtained by

$$\mathbf{z} = \mathbf{BN}(\mathbf{W}_1 \mathbf{x}_i + \mathbf{b}_1) \tag{2}$$

where $\mathbf{W}_1 \in \mathbb{R}^{C_o \times C_i}$ and $\mathbf{b}_1 \in \mathbb{R}^{C_o \times 1}$ are the learnable weights and biases, respectively. Let $\mathbf{z}_t$ be the vector of the key matrix corresponding to the target pixel. Then, the 1-D weight vector of size mn is calculated as follows:

$$\mathbf{w} = \mathbf{SoftMax}\left(\frac{\mathbf{z}^T \mathbf{z}_t}{\sqrt{C_o}}\right) \tag{3}$$

where $\mathbf{SoftMax}(\cdot)$ is the softmax activation function. The scaling factor of $(1/\sqrt{C_o})$ is analogous to that of the self-attention in [38]. The output is $\mathbf{p} \in \mathbb{R}^{C_o}$ obtained by

$$\mathbf{p} = \mathbf{y}\mathbf{w}. \tag{4}$$

Based on the above equations, the features of the target pixel from its one area input are actually the weighted pooling of the value matrix, and the weights are calculated from the correlations between the keys of the target pixel and the other

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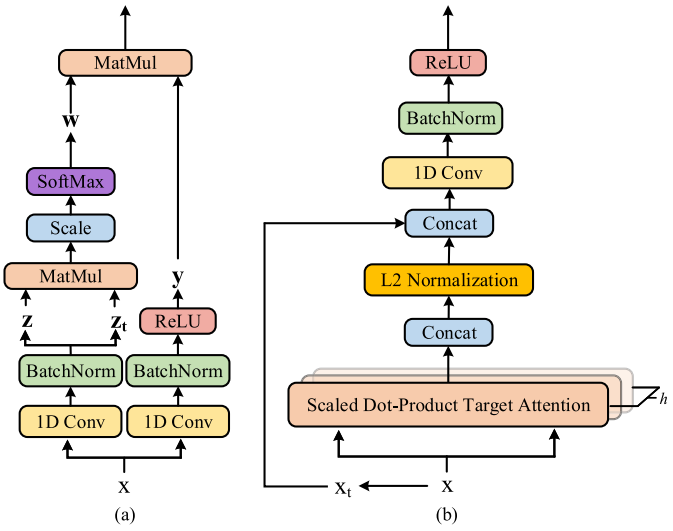


Fig. 5. MSTAMs for feature extraction of target pixels and their similar surrounding pixels of multiscale areas. (a) Scaled dot-product target attention. (b) Multihead target attention extended from scaled dot-product target attention.

pixels in the input. Therefore, there are some differences between MSTAM and self-attention modules. First, position embeddings are generally used in the self-attention module, while they are not used in MSTAM. Second, the self-attention module contains value, query, and key, and the weight matrix is a function of query and key [38]; however, the MSTAM only contains value and key, and the weight vector is a function of keys and the target key. Third, correlation weights between all the pixels are calculated in the self-attention module, while only the correlation weights between the target pixel of an input tensor and its surrounding pixels are calculated in MSTAM.

In practical implementation, the calculations of all the multiarea inputs corresponding to a target pixel are parallel. We calculate all the values and weights of pixels of the largest image patch based on (2)–(4). To calculate the features $\mathbf{p}_j$ of one area input, we fetch the corresponding values $\mathbf{y}_j$ and weights $\mathbf{w}_j$ as

$$\mathbf{p}_j = \frac{\mathbf{y}_j \mathbf{w}_j}{\sum_k w_{jk}} \tag{5}$$

where w_{jk} is the kth element of the weight vector $\mathbf{w}_j$.

2) Multihead Target Attention: As it has been already shown in the self-attention module in Transformer, using multihead attention can help extract more abundant information and achieve better classification performance. Therefore, multihead target attention is designed.

Let the number of heads be h. Then, the value matrix $\mathbf{y} \in \mathbb{R}^{C_o \times mn}$ is divided along the channel dimension into $\mathbf{y}^1, \mathbf{y}^2 \dots \mathbf{y}^h \in \mathbb{R}^{(C_o/h)\times mn}$, and the key matrix $\mathbf{z} \in \mathbb{R}^{C_o \times mn}$ is divided along the channel dimension into $\mathbf{z}^1, \mathbf{z}^2, \dots \mathbf{z}^h \in \mathbb{R}^{(C_o/h)\times mn}$. By (2) and (3), $\mathbf{p}^k$ corresponding to $(\mathbf{y}^k, \mathbf{z}^k)$ in (4) for each k is obtained. The output is

$$\mathbf{p} = \mathbf{Concat}(\mathbf{p}^1, \dots \mathbf{p}^h) \tag{6}$$

where **Concat** is the function that concatenates all the inputs along the channel dimension. Compared with multihead attention in Transformer, the proposed multihead target attention is generalized trivially from the single-head target attention module and no extra parameter is needed.

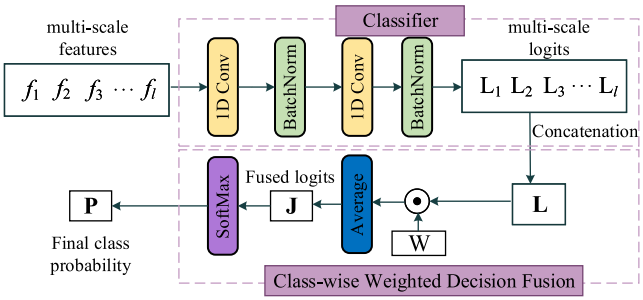


Fig. 6. Classifier and classwise WDF process of the proposed MATA.

3) Multiscale Feature Normalization: In the target attention module, the features of an area are the weighted average of the values corresponding to the area. However, the numbers of the pixels may differ in different areas, and thus, features of different areas have different variations. As a result, there is the feature discrepancy between different areas, which is bad for training one same FE. To mitigate this problem, we normalize the features of each area with L_2-normalization. To be specific, suppose the number of areas is l and the corresponding features calculated from (5) are $\{\mathbf{p}_1, \dots, \mathbf{p}_l\}$, and then the L_2-normalized features are

$$\hat{\mathbf{p}}_j = \frac{\mathbf{p}_j}{\sqrt{\sum_k p_{jk}^2}}, \quad \text{for } 1 \le j \le l \tag{7}$$

where p_{jk} is the kth element of $\mathbf{p}_j$. This is actually inspired from our previous work [39], in which better classification can be achieved when the spectral features are L_2-normalized. We found L_2-normalization is also empirically good in narrowing the discrepancy between areas and serves to aid in achieving better classification performance.

4) Skip Connection: To fully use the spectral information of a target pixel, skip connection is used to concatenate its spectral features $\mathbf{X}_t$ and its spectral–spatial features $\hat{\mathbf{p}}_j$ for the jth area. Then, the concatenated features are passed into 1-D convolutional layer and the output $\tilde{\mathbf{p}}_j$ for the jth area is

$$\tilde{\mathbf{p}}_j = \mathbf{ReLU}(\mathbf{BN}(\mathbf{W}_2\mathbf{Concat}(\hat{\mathbf{p}}_j, \mathbf{X_t}) + \mathbf{b}_2)) \tag{8}$$

where $\mathbf{W}_2 \in \mathbb{R}^{C_o \times (C_o + C_i)}$ and $\mathbf{b}_2 \in \mathbb{R}^{C_o \times 1}$ are the learnable weights and biases, respectively.

C. Multiscale Decision Fusion

Our multiscale decision fusion process is illustrated in Fig. 6, including a parameter-shared classifier and classwise WDF.

1) Parameter-Shared Classifier: After the features of areas have been extracted and L_2-normalized, they are passed into the classifier, which projects the multiscale features into decision space. The classifier contains two linear convolution layers. In the first layer, the outputs with the multiscale features $\{\tilde{\mathbf{p}}_1, \dots \tilde{\mathbf{p}}_l\}$ as the inputs are

$$\mathbf{q}_j = \mathbf{BN}(\mathbf{W}_3\tilde{\mathbf{p}}_j + \mathbf{b}_3), \quad \text{for } 1 \le j \le l \tag{9}$$

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where $\mathbf{W}_3 \in \mathbb{R}^{C_o \times C_o}$ and $\mathbf{b}_3 \in \mathbb{R}^{C_o \times 1}$ are the learnable weights and biases, respectively. It is noted that the parameters of the batch normalization used in (9) are shared across the multiscale features and thus are updated according to all the multiscale features of a batch in the training process. As mentioned earlier, the use of batch normalization can help avoid internal covariate shift. Therefore, the feature discrepancy can be narrowed further by the first layer of the classifier.

In the second layer, the multiscale features in (9) are transformed into logits, i.e.,

$$\mathbf{L}_j = \mathbf{BN}(\mathbf{W}_4 \mathbf{q}_j + \mathbf{b}_4), \quad \text{for } 1 \le j \le l \tag{10}$$

where $\mathbf{W}_4 \in \mathbb{R}^{C \times C_0}$ and $\mathbf{b}_4 \in \mathbb{R}^{C \times 1}$ are the learnable weights and biases, and C is the number of classes. $\mathbf{L}_j \in \mathbb{R}^{C \times 1}$ is the logit corresponding to $\mathbf{q}_j$. By concatenating the multiscale logits, the logit matrix $\mathbf{L} \in \mathbb{R}^{C \times l}$ is obtained by

$$\mathbf{L} = \mathbf{Concat}(\mathbf{L}_1, \ldots \mathbf{L}_l). \tag{11}$$

2) Classwise WDF: After the matrix of multiscale logits is derived by (9)–(11), we average them with per-class weight vectors and obtain the fused logit vector $\mathbf{J} \in \mathbb{R}^{C \times 1}$ as

$$\mathbf{J} = \mathbf{mean}(\mathbf{W} \odot \mathbf{L}, \dim = 1) \tag{12}$$

where **mean** is the average function, and dim = 1 means that the average is conducted along the second dimension, i.e., the dimension corresponding to the scale. $\odot$ is the componentwise product where elements of two matrices in the same position are multiplied. $\mathbf{W} \in \mathbb{R}^{C \times l}$ is the weight matrix

$$\mathbf{W} = \mathbf{SoftMax}(\mathbf{U}, \dim = 1) \tag{13}$$

where $\mathbf{U} \in \mathbb{R}^{C \times l}$ is a learnable matrix and **Softmax** is conducted along the dimension corresponding to the scale. The final probability vector $\mathbf{P}$ is

$$\mathbf{P} = \mathbf{SoftMax}(\mathbf{J}). \tag{14}$$

D. Training Objective

In many multiscale models, the FEs and/or classifiers of different scales are trained separately (e.g., DRCNN [17], SDNet [36]), consuming a substantial amount of training time. Since our model has only one shared FE and one shared classifier, it is trained with the multiscale area inputs simultaneously. Based on the probability vector in (14), the training loss is

$$\text{Loss} = -\frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{C} \log(P_{ij}) \mathbf{1}(\text{label}_i = j) \tag{15}$$

where P_{ij} is the probability of sample i being class j, and $\mathbf{1}$ is a logic function that outputs 1 if the input is true, or 0 if the input is false.

IV. Experimental Results and Analysis

This section is devoted to a comprehensive analysis of the proposed MATA to demonstrate its effectiveness. To begin with, the experimental datasets including Indian Pines (IP), MUFFL, Gaofeng State Owned Forest Farm (GFSOFF), Houston2013, and Houston2018 are introduced. Then, the impact of the hyperparameters, namely, the largest patch size R, the number of heads, and the embedding dimension of each head, is investigated on these datasets. Experiments are carried out in comparison to several state-of-the-art models to evaluate the classification performance of MATA. To gain insights into shared FE and classifier, and the individual components of MATA, ablation experiments are conducted to compare with MATA of nonshared FE and classifier and evaluate the effectiveness of the multiscale feature extraction strategy, L_2-normalization of the multiscale feature, and classwise WDF strategy. Furthermore, the classification performance of MATA under different training samples compared with other models is investigated. Finally, the computational complexity is also studied.

TABLE I
Class Labels and Train–Test Distribution of Samples for IP

Class No.	Class Name	Train	Test
1	Alfalfa	23	23
2	Corn-notil	100	1328
3	Corn-mintill	100	730
4	Corn	100	137
5	Grass-pasture	100	383
6	Grass-trees	100	630
7	Grass-pasture-mowed	14	14
8	Hay-windrowed	100	378
9	Oats	10	10
10	Soybean-notill	100	872
11	Soybean- mintill	100	2355
12	Soybean-clean	100	493
13	Wheat	100	105
14	Woods	100	1165
15	Buildings-Grass-Trees-Drives	100	286
16	Stone-Steel-Towers	47	46
	Total	1294	8955

A. Experimental Datasets

1) Indian Pines: The first dataset is the IP dataset, which was taken by the airborne visible/infrared imaging spectrometer AVIRIS sensor over Indiana. It consists of 145 × 145 pixels with a spatial resolution of 20 × 20 m and 200 spectral bands in the wavelength range of 0.4–1.0 μm bands after removing the spectra covering the water absorption range. It contains 16 different crop classes, and the numbers of training and testing samples of each class are listed in Table I. The false-color image of IP and the distribution of those samples are shown in Fig. 7.

2) MUFFL: The second dataset is the MUFFL dataset, which was collected by CASI-1500 sensor over the University of Southern Mississippi in November 2010. It consists of 325 × 220 pixels with a spatial resolution of 0.54 × 1 m and 64 spectral bands in the wavelength range of 0.38–1.05 μm. It contains 11 different classes, and the numbers of training and testing samples of each class are listed in Table II. The false-color image of MUFFL and the distribution of those samples are shown in Fig. 7.

3) Gaofeng State Owned Forest Farm: The third HSI dataset is the GFSOFF dataset, which was obtained by the AISA Eagle II diffraction grating push-broom hyperspectral imager carried by the airborne LiCHy (LiDAR, CCD, and

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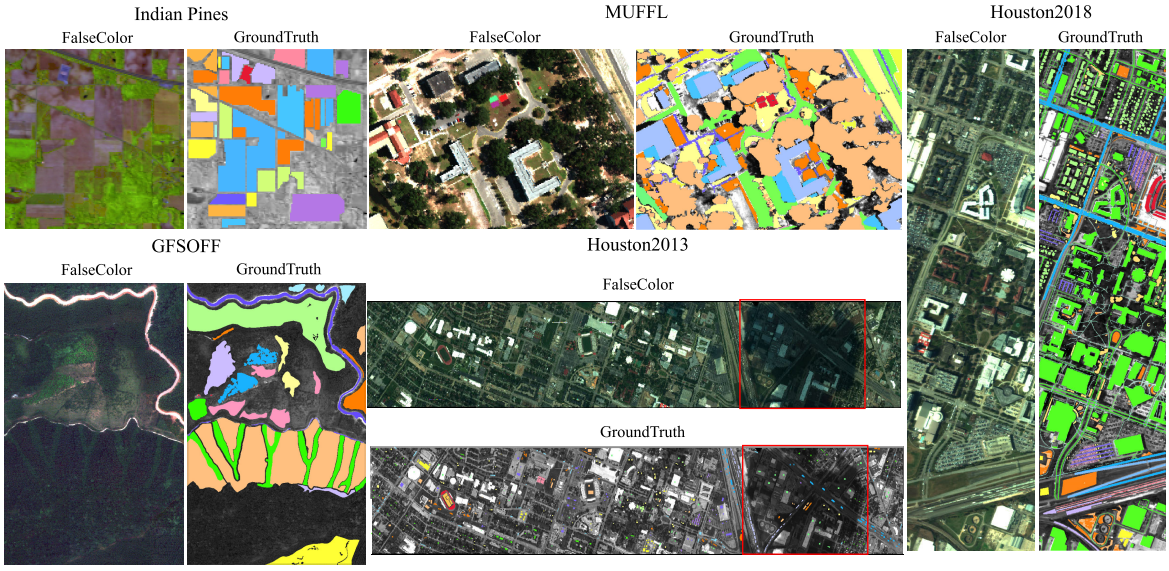


Fig. 7. False-color images and the distributions of training and testing samples for the IP, MUFFL, GFSOFF, Houston2013, and Houston2018 datasets, respectively.

TABLE II
CLASS LABELS AND TRAIN–TEST DISTRIBUTION OF SAMPLES FOR MUFFL

No.	Class Name	Train	Test
1	Trees	150	23096
2	Mostly grass	150	4120
3	Mixed ground surface	150	6732
4	Dirt and sand	150	1676
5	Road	150	6537
6	Water	150	316
7	Building shadow	150	2083
8	Building	150	6090
9	Sidewalk	150	1235
10	Yellow curb	150	33
11	Cloth panels	150	119
	Total	1650	52037

TABLE III
CLASS LABELS AND TRAIN–TEST DISTRIBUTION OF SAMPLES FOR GFSOFF

Class No.	Class Name	Train	Test
1	Cuninghamia lanceolata	4038	76705
2	Pinus massoniana	268	5072
3	Pius elliottii	208	3967
4	Eucalyptus grandis x urophylla	746	14185
5	Eucalyptus urophylla	2014	38261
6	Catanopsis hystrix	1274	24187
7	Mytilaria laosensis	150	2850
8	Camellia oleifera	416	7908
9	Other broadleaf forest	538	10203
10	Road	524	9948
11	Cutting blank	508	9662
12	Building land	4	64
	Total	10688	203012

Hyperspectral) system of the Chinese Academy of Forestry Sciences over GFSOFF in Guangxi province in South China in January 2018 [22]. It consists of 572 × 906 pixels with a spatial resolution of 1 × 1 m and 125 spectral bands in the wavelength range of 0.4–1.0 μm. It contains nine different forest vegetation classes and three nonforest vegetation classes, and the numbers of training and testing samples of each class are listed in Table III. The false-color image of GFSOFF and the distribution of those samples are shown in Fig. 7.

4) Houston2013: The fourth HSI dataset is the Houston2013 dataset, which was gathered by the CASI-1500 senor over the University of Houston and neighboring areas in June 2012 for the 2013 GRSS Data Fusion Contest [40]. It consists of 349 × 1905 pixels with a spatial resolution of 2.5 m and 144 spectral bands in the wavelength range of 0.38–1.05 μm. It contains five natural objects and ten man-made objects, and the numbers of training and testing samples of each class are listed in Table IV. The false-color image of Houston2013 and the distribution of those samples are shown in Fig. 7. Note that in the red rectangle of the false-color image of Houston2013, there is an area affected by cloud.

5) Houston2018: The fifth HSI dataset is the Houston2018, which was captured by the ITRES CASI-1500 senor over the University of Houston campus in the February of 2017 [41]. It consists of 601 × 2384 pixels with a spatial resolution of 1 m and 48 spectral bands in the wavelength range of 0.38–1.05 μm. It contains 20 classes, and the numbers of training and testing samples of each class are listed in Table V. The false-color image of Houston2018 and the distribution of those samples are shown in Fig. 7.

B. Hyperparameter Study

One important module of the proposed MATA is MSTAM, which has two hyperparameters, i.e., the number of heads and the embedding dimension of each head. Therefore, we study the classification performance by the metric of overall accuracy (OA) for convenience under different settings of the two hyperparameters. By fixing the input HSI patch sizes, we choose the values for the number of heads and the embedding dimension of each head from the sets {4, 6, 8, 10, 12} and {8, 16, 32, 64}, respectively. As shown in Fig. 8,

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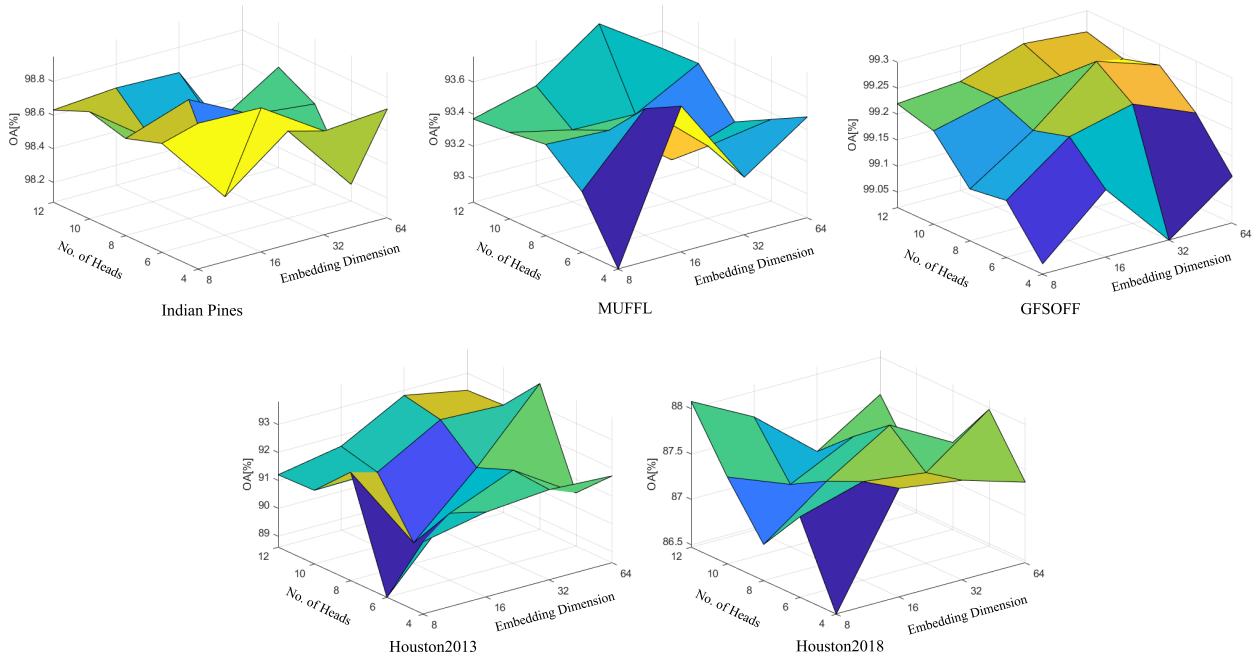


Fig. 8. OAs[%] versus the embedding dimension and number of heads of the MSTA for the IP, MUFFL, GFSOFF, Houston2013, and Houston2018 datasets, respectively.

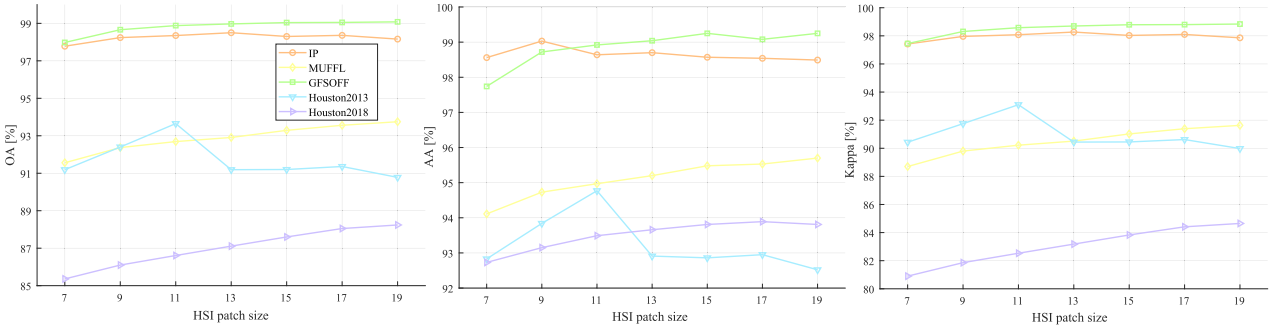


Fig. 9. OA [%], AA [%], and Kappa [%] versus the largest HSI patch size R on the IP, MUFFL, GFSOFF, Houston2013, and Houston2018 datasets for the proposed MATA.

TABLE IV
CLASS LABELS AND TRAIN–TEST DISTRIBUTION OF SAMPLES FOR HOUSTON2013

Class No.	Class Name	Train	Test
1	Grass healthy	198	1053
2	Grass stressed	190	1064
3	Grass synthetics	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot 1	192	1041
13	Parking lot 2	184	285
14	Tennis court	181	247
15	Running track	187	473
Total		2832	12197

TABLE V
CLASS LABELS AND TRAIN–TEST DISTRIBUTION OF SAMPLES FOR HOUSTON2018

No.	Class Name	Train	Test
1	Healthy grass	200	9599
2	Stressed grass	200	32302
3	Artificial turf	200	484
4	Evergreen trees	200	13395
5	Deciduous trees	200	4821
6	Bare earth	200	4316
7	Water	133	133
8	Residential buildings	200	39572
9	Non-residential buildings	200	223552
10	Roads	200	45666
11	Sidewalks	200	33829
12	Crosswalks	200	1318
13	Major thoroughfares	200	46148
14	Highways	200	9665
15	Railways	200	6737
16	Paved parking lots	200	11300
17	Unpaved parking lots	68	68
18	Cars	200	6347
19	Trains	200	5169
20	Stadium seats	200	6424
Total		3806	501050

the OAs on the five datasets vary over different settings of the number of heads and the embedding dimension of each head, but the degrees of variation are not significant. For convenience, we choose a typical setting, i.e., 8 for the number of heads and 64 for the embedding dimension of each head, for the rest of the experiments.

Furthermore, the effect of the largest image patch size on the classification performance is analyzed experimentally. The

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TABLE VI
CLASSIFICATION PERFORMANCE [%] OF DIFFERENT MODELS ON THE IP DATASET

Class No.	RF [42]	SVM [43]	CNNHSI [18]	SSCNN [21]	3DCNN [22]	BERTHSI [23]	CDCNN [30]	DRCNN [17]	FusCNNGCN [33]	PMCN [34]	MATA
1	82.61	86.96	73.91	21.74	**100**	**100**	91.30	**100**	91.30	**100**	**100**
2	57.98	74.47	71.84	17.32	90.96	91.42	91.42	97.29	85.02	96.39	**97.74**
3	74.11	80.27	64.79	54.93	98.08	97.67	94.79	97.67	92.47	**99.32**	98.77
4	89.05	83.21	95.62	89.05	98.54	**100**	**100**	**100**	95.62	**100**	98.54
5	86.95	93.73	91.91	91.12	98.17	94.52	92.95	**100**	95.30	98.96	98.43
6	96.03	96.35	90.95	96.51	**100**	96.98	98.10	**100**	99.37	99.52	**100**
7	85.71	85.71	85.71	50.00	**100**	**100**	**100**	**100**	85.71	**100**	85.71
8	98.15	98.68	99.74	97.35	**100**	**100**	**100**	**100**	**100**	**100**	**100**
9	80.00	**100**	80.00	0.00	**100**	**100**	0.00	**100**	90.00	**100**	**100**
10	73.05	73.85	26.61	78.67	91.97	96.67	93.35	91.06	88.19	**99.31**	96.33
11	65.48	69.21	48.87	65.01	86.28	94.86	75.84	89.00	74.78	95.97	**98.17**
12	73.63	81.14	86.21	58.42	91.89	94.12	98.17	95.54	91.68	97.97	**98.78**
13	99.05	98.10	99.05	96.19	99.05	**100**	98.10	**100**	99.05	**100**	99.05
14	87.64	88.07	80.34	94.16	95.19	98.11	90.30	96.22	95.45	99.74	**99.91**
15	77.27	74.83	80.07	54.20	98.60	**100**	99.30	**100**	96.85	**100**	99.65
16	**100**	**100**	91.30	**100**	**100**	**100**	97.83	**100**	**100**	**100**	**100**
OA	74.99	79.65	67.18	66.93	92.89	95.85	89.31	94.91	87.85	98.06	**98.51**
AA	82.92	86.54	79.18	66.54	96.80	97.77	88.84	97.92	92.55	**99.20**	98.19
Kappa	71.53	76.73	62.90	62.00	91.80	95.19	87.74	94.12	86.09	97.75	**98.28**

largest image patch size R is chosen from the set {7, 9, 11, 13, 15, 17, 19}. For the training setting, we use the ADAM optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$, and the learning rate is decayed every 20 epochs with a decay factor of $\gamma = 0.6$. In the subsequent experiments, the initial learning rate was set to 0.005 for the IP, MUFFL, GFSOFF, and Houston2018 datasets and 0.001 for the Houston2013 dataset. The batch size is set as 64 for all the datasets. In addition to the metric OA, average accuracy (AA) and Kappa coefficient are used to evaluate the classification performance. As shown in Fig. 9, the overall trend of classification accuracy increases with increase in the largest image patch size on all the datasets except for the IP and Houston2013 datasets. This indicates that the proposed model can extract effective spatial information and achieve better performance with the increase in spatial size. For the IP dataset, the classification performance on the largest image patch size is comparable, achieving the best at patch size 13. For the Houston2013 dataset, the classification accuracy increases with the increase in the largest image patch size and then decreases, and its best patch size is 11. To balance the computational cost and classification accuracy, the largest image patch size is set as 11 for the Houston2013, 13 for the IP, and 15 for the MUFFL, GFSOFF and Houston2018 datasets in all the subsequent experiments.

C. Classification Performance

To validate the effectiveness of the proposed MATA, experiments are conducted on the above-mentioned five datasets in comparison to the state-of-the-art models, including traditional classification models, i.e., random forest (RF [42]) and support vector machine (SVM [43]), single-scale CNN models, i.e., CNNHSI [18], SSCNN [21], 3DCNN [22], and a single-scale transfomer-like model, i.e., BERT-HSI [23], and multiscale models, i.e., CDCNN [30], DRCNN [17], FusCNNGCN [33], and PMCN [34]. The hyperparameters of these methods are set by their recommendations. The data preprocessing is important for HSI classification, and normalization methods including standardization, 0–1 normalization, and LN [39] are tried to use on the five datasets to find the suitable one for all the comparison methods. The class-specific accuracy (CA), OA, AA, and Kappa coefficient are used to evaluate the classification performance.

1) Classification Accuracy: Classification performances of different methods on IP, MUFFL, GFSOFF, Houston2013, and Houston2018 are listed in Tables VI–X, respectively. The best results are shown in boldface. For all the experiments on the five datasets, the baselines produced by the traditional methods RF and SVM are generally worse than other deep-learning-based methods excerpt for a few methods. The classification performance of the single-scale CNN model CNNHSI is generally worse than the other single-scale ones except on the Houston2013 dataset where 3DCNN performs the worst. This is because only spectral information is used in CNNHSI. By comparison, the single-scale transformer-like model BERTHSI achieves generally better classification performance than the other single-scale models. This is because it can extract more effective spectral–spatial information than CNN models by equipping more powerful self-attention modules.

For multiscale models, they perform better than the single-scale models on average. CDCNN performs consistently worse than the other multiscale models on all the datasets. This is because CDCNN uses a multiscale filter bank only at the very beginning of the feature extraction and cannot use multiscale features effectively. By comparison, PMCN achieves better classification performance than CDCNN as it uses multiscale filter bank in many stages of feature extraction. For the multiscale models using multiple feature extraction branches, FusCNNGCN achieves better classification performance than all the other comparison models. Using different feature extraction branches for different areas, the multiscale model DRCNN performs stably well on all the datasets, and it is not the best nor the worst comparison model.

The performance of the proposed MATA is superior and better than all the other aforementioned state-of-the-art models

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TABLE VII
CLASSIFICATION PERFORMANCE [%] OF DIFFERENT MODELS ON THE MUFFL DATASET

Class No.	RF [42]	SVM [43]	CNNHSI [18]	SSCNN [21]	3DCNN [22]	BERTHSI [23]	CDCNN [30]	DRCNN [17]	FusCNNGCN [33]	PMCN [34]	MATA
1	82.31	81.17	81.88	89.50	87.70	82.72	87.17	89.98	88.29	92.09	**94.40**
2	79.78	77.50	83.01	69.73	80.32	78.01	72.99	86.04	79.51	85.12	**88.30**
3	68.55	68.36	72.12	75.19	76.63	76.98	65.63	81.74	77.09	90.00	**91.64**
4	78.76	77.45	89.86	93.20	87.41	91.95	81.68	93.91	93.50	95.11	**97.73**
5	85.53	84.79	84.41	85.93	82.44	85.25	92.29	91.22	91.08	89.44	**93.18**
6	92.72	90.51	97.78	**100**	99.37	99.68	99.37	99.37	99.05	99.05	98.73
7	86.13	84.01	77.82	86.80	90.78	94.29	94.14	**97.50**	95.20	95.01	95.30
8	75.48	77.64	67.78	84.14	82.35	78.51	82.92	90.94	87.14	95.11	**96.04**
9	67.94	71.74	62.11	72.87	82.11	89.39	**95.79**	87.37	86.07	87.53	92.31
10	81.82	84.85	96.97	93.94	90.91	87.88	96.97	93.94	**100**	90.91	**100**
11	97.48	92.44	99.16	99.16	98.32	99.16	98.32	99.16	**100**	99.16	**100**
OA	79.73	79.12	79.15	84.71	84.48	82.49	83.82	89.32	86.85	91.45	**93.73**
AA	81.50	80.95	82.99	86.41	87.12	87.62	87.93	91.92	90.63	92.59	**95.24**
Kappa	73.94	73.24	73.20	80.09	79.82	77.51	79.14	86.04	82.87	88.77	**91.61**

TABLE VIII
CLASSIFICATION PERFORMANCE [%] OF DIFFERENT MODELS ON THE GFSOFF DATASET

Class No.	RF [42]	SVM [43]	CNNHSI [18]	SSCNN [21]	3DCNN [22]	BERTHSI [23]	CDCNN [30]	DRCNN [17]	FusCNNGCN [33]	PMCN [34]	MATA
1	90.21	85.80	82.53	91.61	95.43	96.99	92.05	98.00	95.78	**99.87**	99.34
2	18.91	10.13	83.55	75.51	92.84	92.94	77.43	92.61	89.33	98.03	**99.59**
3	4.44	7.64	99.60	85.53	97.08	93.80	73.56	92.74	84.93	94.71	**99.87**
4	58.88	48.68	98.58	86.07	96.84	97.46	91.62	99.28	98.06	99.53	**99.93**
5	70.93	55.89	98.67	96.73	97.51	92.44	99.09	99.58	98.48	99.91	**99.93**
6	39.14	31.20	**99.30**	85.07	84.41	90.09	76.52	86.41	83.82	83.40	96.84
7	0.32	8.39	71.08	70.28	94.28	83.58	73.86	96.91	92.67	99.93	**100**
8	57.78	43.11	62.58	95.52	98.47	99.97	94.92	99.06	97.77	**100**	99.89
9	42.87	37.45	59.87	68.81	90.74	93.46	77.59	89.16	83.40	96.07	**99.47**
10	97.32	92.67	96.24	99.10	99.01	99.69	99.66	**99.76**	99.49	99.25	99.43
11	95.55	84.72	81.69	99.15	99.69	98.34	99.22	99.83	99.81	**99.88**	99.74
12	0.00	0.00	61.19	90.63	45.31	68.75	65.63	26.56	48.44	78.13	**100**
OA	70.52	62.72	59.30	90.32	94.80	95.12	90.60	96.50	94.42	97.52	**99.27**
AA	48.03	42.14	**100**	87.00	90.97	92.29	85.09	89.99	89.33	95.72	**99.50**
Kappa	61.41	51.10	98.10	87.79	93.45	93.84	88.11	95.57	92.95	96.85	**99.08**

TABLE IX
CLASSIFICATION PERFORMANCE [%] OF DIFFERENT MODELS ON THE HOUSTON2013 DATASET

Class No.	RF [42]	SVM [43]	CNNHSI [18]	SSCNN [21]	3DCNN [22]	BERTHSI [23]	CDCNN [30]	DRCNN [17]	FusCNNGCN [33]	PMCN [34]	MATA
1	82.81	83.10	82.53	82.53	80.63	83.00	82.43	82.34	82.62	82.43	**91.64**
2	82.80	83.18	83.55	**84.96**	**84.96**	83.36	**84.96**	84.77	78.10	**84.96**	83.83
3	98.61	99.80	99.60	98.81	95.84	91.09	**100**	88.71	91.68	98.22	**100**
4	89.77	97.16	98.58	92.23	71.40	93.56	93.37	92.33	96.31	92.14	**99.24**
5	96.59	99.05	98.67	99.24	90.06	99.81	**100**	99.91	**100**	**100**	99.34
6	99.30	99.30	99.30	93.01	**100**	98.60	95.80	99.30	98.60	**100**	**100**
7	76.03	85.35	71.08	94.12	**93.56**	87.22	79.38	89.09	91.23	89.55	85.91
8	38.56	47.48	62.58	68.95	85.57	68.76	71.23	67.90	85.85	74.26	**94.59**
9	68.93	74.88	59.87	80.64	72.71	77.15	78.75	78.00	72.71	81.40	**91.78**
10	51.35	53.96	96.24	43.24	42.57	48.46	49.71	63.51	61.49	56.47	**91.12**
11	66.79	79.70	81.69	59.49	65.65	82.73	79.22	72.68	79.51	79.41	**97.91**
12	53.99	76.46	61.19	90.11	78.48	86.17	82.54	83.39	92.03	92.60	**95.39**
13	64.91	70.53	59.30	85.61	67.02	86.32	92.63	89.47	90.53	71.58	**92.98**
14	97.57	99.60	**100**	95.95	99.19	99.60	**100**	**100**	99.60	**100**	**100**
15	95.35	97.46	98.10	94.50	84.36	98.94	77.37	83.09	90.70	82.88	**100**
OA	73.68	80.26	81.30	81.63	78.25	82.90	82.32	83.28	85.23	84.23	**93.83**
AA	77.56	83.13	83.49	84.23	80.80	85.65	85.73	85.52	87.40	85.73	**94.92**
Kappa	71.66	78.64	79.69	80.05	76.49	81.51	80.95	81.94	83.96	82.95	**93.31**

by approximately 0.5% on the IP dataset, 1.7% on the MUFFL dataset, 1.5% on the GFSOFF dataset, 8.6% on the Houston2013 dataset, and 3% on the Houston2018 dataset over the second highest other methods in terms of OA. This indicates that by equipping with the proposed MSTAM and classwise decision weighting strategy, MATA can extract more effective spectral–spatial features and make better classification decisions.

2) Visualization: To visually demonstrate the effectiveness of the proposed MATA, classification maps of all the

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TABLE X
CLASSIFICATION PERFORMANCE [%] OF DIFFERENT MODELS ON THE HOUSTON2018 DATASET

Class No.	RF [42]	SVM [43]	CNNHSI [18]	SSCNN [21]	3DCNN [22]	BERTHSI [23]	CDCNN [30]	DRCNN [17]	FusCNNGCN [33]	PMCN [34]	MATA
1	95.99	**98.59**	99.77	89.68	95.47	89.51	95.68	94.72	97.35	95.56	94.58
2	90.68	92.92	72.56	90.57	90.44	91.60	90.95	91.18	89.36	90.63	**94.50**
3	**100**	**100**	**100**	99.17	98.76	**100**	**100**	**100**	**100**	**100**	**100**
4	94.98	95.75	92.21	96.13	96.86	97.85	96.68	97.81	96.10	**98.27**	98.01
5	79.57	87.14	77.95	88.90	88.05	95.93	91.74	97.16	92.57	95.89	**98.92**
6	94.86	98.52	91.03	99.17	98.29	99.49	98.98	99.79	99.42	99.98	**100**
7	99.25	**100**	99.25	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
8	71.48	75.07	75.26	85.71	71.30	83.85	77.86	86.93	80.16	**92.72**	90.22
9	56.90	70.48	51.85	69.17	67.46	79.68	81.96	85.65	82.68	85.13	**89.43**
10	33.50	41.79	4.09	43.57	51.99	60.48	**70.85**	58.63	49.90	66.78	70.56
11	37.54	47.35	31.17	53.79	58.85	62.56	70.95	67.90	60.53	69.30	**74.95**
12	56.98	60.09	61.84	40.67	64.95	78.91	81.56	76.63	70.86	**90.67**	85.58
13	37.16	40.56	43.26	64.17	54.90	64.99	51.68	74.05	63.35	78.03	**82.99**
14	79.24	77.63	70.08	78.07	95.39	97.03	79.68	97.90	94.88	**99.28**	98.76
15	95.49	93.94	87.00	96.70	97.02	98.46	95.56	99.38	98.99	**99.61**	99.58
16	85.65	88.25	84.80	97.15	92.19	93.25	95.42	96.81	92.77	97.06	**97.27**
17	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
18	72.77	77.31	69.95	86.21	90.29	96.66	96.25	96.91	92.85	**97.86**	96.91
19	83.34	87.75	76.94	95.51	96.77	98.70	91.20	99.57	97.60	99.65	**99.75**
20	89.78	95.79	99.91	99.98	98.61	98.79	97.92	**100**	98.91	99.98	99.95
OA	59.85	68.44	53.86	71.62	70.42	79.11	79.52	83.26	78.70	84.75	**87.88**
AA	77.76	81.45	74.45	83.72	85.38	89.39	88.25	91.05	87.92	92.82	**93.60**
Kappa	52.72	61.48	46.44	65.48	64.14	73.96	74.31	78.83	73.27	80.75	**84.54**

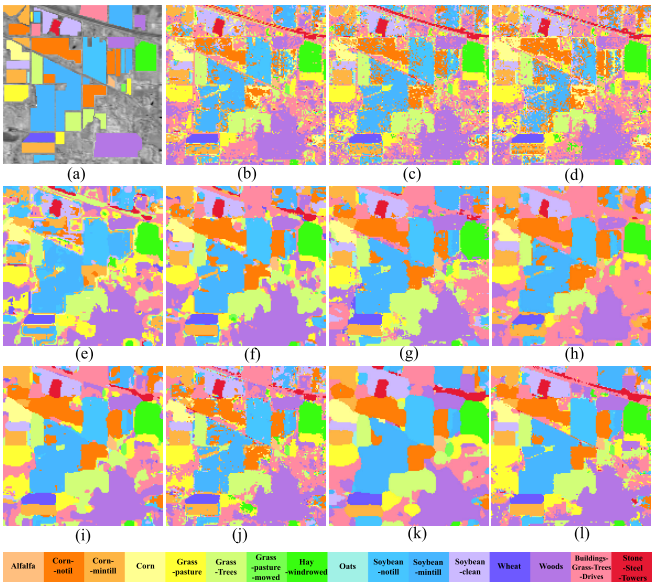


Fig. 10. Classification maps of the IP dataset produced by different models. (a) Ground truth. (b) RF (74.99%). (c) SVM (79.65%). (d) CNNHSI (67.18%). (e) SSCNN (66.93%). (f) 3DCNN (92.89%). (g) BERTHSI (95.85%). (h) CDCNN (89.31%). (i) DRCNN (94.91%). (j) FusCNNGCN (87.85%). (k) PMCN (98.06%). (l) MATA (98.51%).

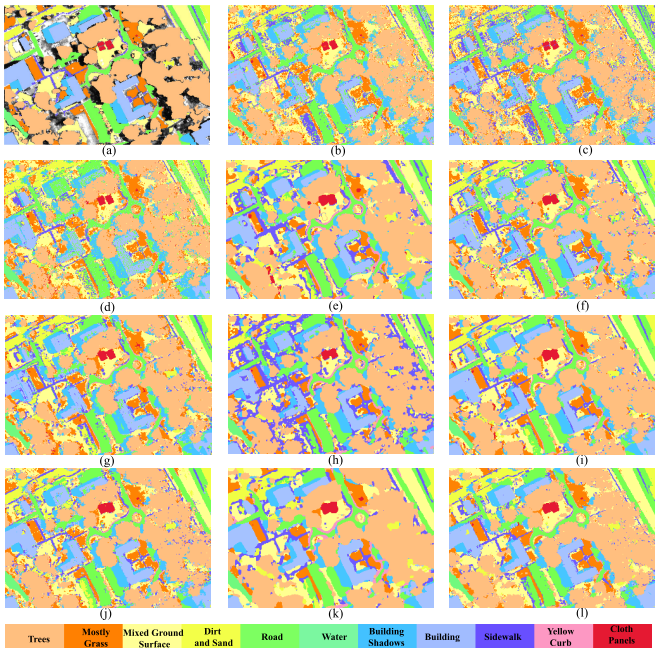


Fig. 11. Classification maps of the MUFFL dataset produced by different models. (a) Ground truth. (b) RF (79.73%). (c) SVM (79.12%). (d) CNNHSI (79.15%). (e) SSCNN (84.71%). (f) 3DCNN (84.48%). (g) BERTHSI (84.49%). (h) CDCNN (83.82%). (i) DRCNN (89.32%). (j) FusCNNGCN (86.85%). (k) PMCN (91.45%). (l) MATA (93.73%).

aforementioned models on the three datasets are illustrated in Figs. 10–14. The produced maps by CNNHSI contain many salt-and-pepper pixels, which is consistent with the worse results listed in Tables VI–X. This is because spatial information is not used in CNNHSI and the pixel may be incorrectly classified with only spectral information. However, it produces a better map of highway covered by cloud as many pixels of highway are correctly classified on the Houston2013 dataset. By comparison, the classification performances of PMCN as listed in Tables VI–X are acceptable, but the produced maps are not satisfactory since they are oversmoothed, and many small objects are submerged. For example, in its classification map on the GFOSFF dataset, the building land in the red rectangle has been incorrectly covered by other classes as shown in Fig. 12. In addition, many pixels including pixels of highway in the area covered by cloud are incorrectly classified as running track on the Houston2013 dataset.

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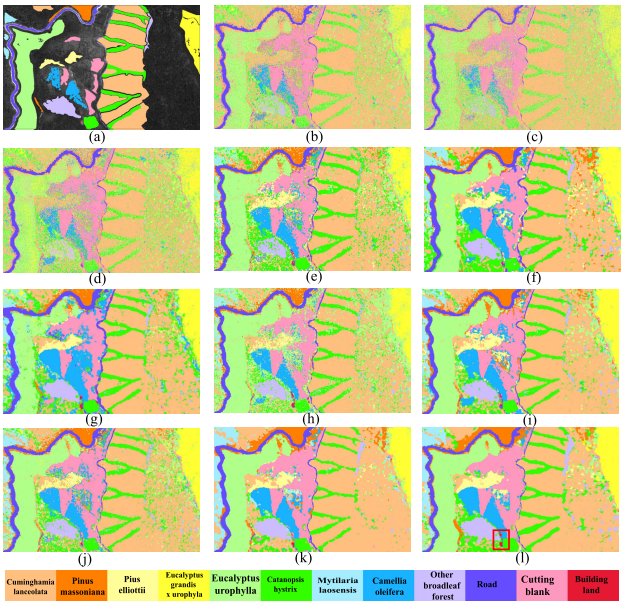


Fig. 12. Classification maps of the GFSOFF dataset produced by different models. (a) Ground truth. (b) RF (70.52%). (c) SVM (62.72%). (d) CNNHSI (59.30%). (e) SSCNN (90.32%). (f) 3DCNN (94.80%). (g) BERTHSI (95.12%). (h) CDCNN (90.60%). (i) DRCNN (96.50%). (j) FusCNNGCN (94.42%). (k) PMCN (97.52%). (l) MATA (99.27%).

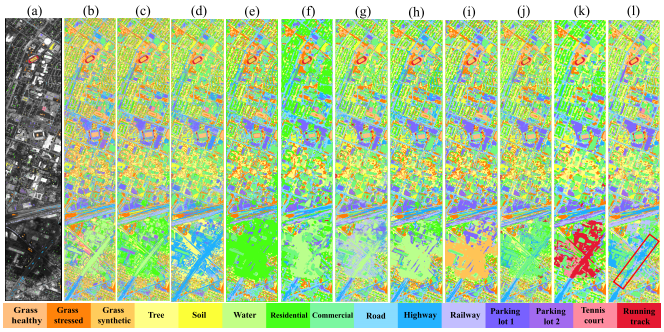


Fig. 13. Classification maps of the Houston2013 dataset produced by different models. (a) Ground truth. (b) RF (73.68%). (c) SVM (80.26%). (d) CNNHSI (83.10%). (e) SSCNN (81.63%). (f) 3DCNN (78.25%). (g) BERTHSI (82.90%). (h) CDCNN (82.32%). (i) DRCNN (83.28%). (j) FusCNNGCN (85.23%). (k) PMCN (84.23%). (l) MATA (93.83%).

However, the proposed MATA produces more accurate and spatially smoother classification maps with less mislabeled pixels than the other models, which are consistent with the results listed in Tables VI–X. In particular, the building land (it is noted that the number of its training samples is only 4) is well-preserved in the maps of GFSOFF produced by MATA, indicating that ground objects with small sizes can be identified correctly by the proposed model, confirming the benefit of feature extraction of multiscale areas. The Houston2013 dataset poses a significant challenge due to the presence of shadows created by cloud, which causes pixels within the cloud-covered area to have different spectral characteristics than those in other areas of the dataset. Moreover, the majority of the training data are collected from noncloud-covered areas, exacerbating this issue. Notably, the state-of-the-art models, apart from CNNHSI, fail to accurately label the highway in the cloud-covered area, as shown in the red rectangle in Fig. 13.

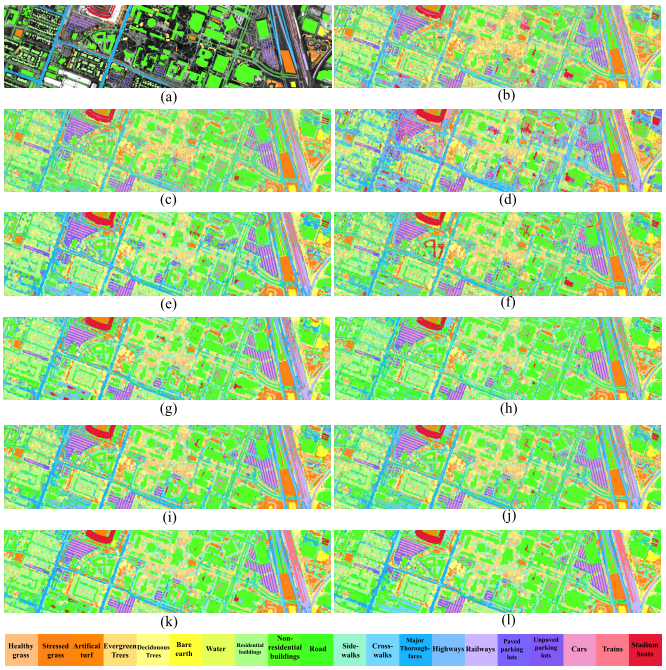


Fig. 14. Classification maps of the Houston2018 dataset produced by different models. (a) Ground truth. (b) RF (59.85%). (c) SVM (68.44%). (d) CNNHSI (53.86%). (e) SSCNN (71.62%). (f) 3DCNN (70.42%). (g) BERTHSI (79.11%). (h) CDCNN (79.52%). (i) DRCNN (83.26%). (j) FusCNNGCN (78.70%). (k) PMCN (84.75%). (l) MATA (87.88%).

In contrast, the proposed MATA successfully classifies the highway due to their ability to extract optimal spectral–spatial information from multiscale areas and correctly give weights to the decisions from these areas. This approach enables the extraction of crucial pixel properties from both cloud-covered and noncloud-covered areas, allowing for discrimination of the pixels within the cloud-covered area to some degree using the information of multiscale areas.

D. Ablation Experiment

In this section, we examine the classification performance when FE and classifier are shared or not in the proposed MATA. In addition, we perform ablation experiments to verify the effectiveness of the important components of the proposed MATA, including the multiscale feature extraction strategy, L_2-normalization, and classwise WDF strategy. To be specific, this is done by introducing and removing these strategies or modules. We use OA, AA, and Kappa coefficient to quantize the classification performance.

1) Shared or Nonshared FE and Classifier: As shown in Table XI, the comparison between the classification performances achieved with nonshared FEs and classifiers and shared counterparts reveals that while the former generally outperforms the latter, the difference is marginal. For instance, on the GFSOFF dataset, the nonshared approach only edges out the shared variant by a mere 0.05% in terms of OA. Conversely, on the Houston2013 dataset, MATA by shared FE and classifier outperforms the nonshared counterpart by 0.3%. It is important to note that these differences in performance are negligible. Other metrics, i.e., AA and Kappa, also exhibit that those models have comparable classification performance.

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TABLE XI
ABLATION STUDY OF SHARING FE AND CLASSIFIER

	Shared FE	Shared Classifier	OA (%)	AA (%)	Kappa (%)	Training Time (s)	Testing Time (s)	Total Parameters (k)
IP	×	×	98.92	99.20	98.74	595.85	6.53	24447.20
	×	✓	98.74	98.92	98.54	478.51	6.03	17687.60
	✓	×	98.60	98.84	98.38	247.88	4.40	7603.94
	✓	✓	98.51	98.19	98.28	128.38	4.10	844.34
MUFFL	×	×	94.14	95.90	92.16	751.67	18.00	17277.79
	×	✓	93.73	95.86	91.60	627.98	15.54	10582.56
	✓	×	93.86	95.81	91.78	391.19	7.50	7328.09
	✓	✓	93.73	95.24	91.61	51.73	3.15	632.87
GFSOFF	×	×	99.34	98.76	99.17	3551.94	81.33	20476.84
	×	✓	99.24	95.17	99.04	1954.53	48.15	13768.74
	✓	×	99.17	99.46	98.95	1084.88	38.64	7435.18
	✓	✓	99.27	99.50	99.08	415.60	42.70	727.08
Houston2013	×	×	93.50	94.85	92.94	884.73	81.31	21509.27
	×	✓	93.72	94.85	93.19	536.32	67.45	14762.54
	✓	×	93.17	94.54	92.58	301.89	59.00	7504.53
	✓	✓	93.83	94.93	93.31	118.90	57.72	757.81
Houston2018	×	×	88.10	94.00	84.48	1189.9	221.08	16562.71
	×	✓	87.53	92.93	83.73	935.22	169.33	9751.61
	✓	×	87.27	93.06	83.38	564.32	87.86	7424.02
	✓	✓	87.88	93.60	84.54	239.88	72.79	612.92

Moreover, from the computational complexity perspective, MATA by the shared FE and classifier significantly reduces both training and testing times across all the five datasets when compared with the nonshared counterparts. In addition, the total parameters of the MATA with nonshared FEs and classifiers are orders of magnitude greater than those of the MATA using shared FEs and classifiers, reaching a difference of 100-fold. Overall, the proposed MATA using shared FE and classifier showcases a classification performance akin to its nonshared counterparts, all the while boasting fewer parameters and considerably shorter training and testing times.

2) Multiscale or Single-Scale Feature Extraction (MSFE): We use the proposed MSTAM for multiscale feature extraction by performing it on multiple sizes and region parts of the HSI patch. For the single-scale feature extraction, we perform it only on the largest HSI patch. As listed in Table XII, the classification performance is improved after introducing our multiscale feature extraction module. When multiscale features are used, the overall classification accuracies on the IP, MUFFL, GFSOFF, Houston2013, and Houston2018 datasets are improved from 95.90%, 91.82%, 98.72 %, 90.97%, and 84.68% to 97.95%, 93.04%, 99.19%, 92.38%, and 86.47%, respectively. In particular, the improvements on the IP, Houston2013 and Houston2018 datasets are significant, approximately 2%, 1.4%, and 1.8%, respectively. Other metrics, i.e., AA and Kappa, exhibit the similar improvement when multiscale feature extraction is introduced. This indicates that our MSTAM enhances the effective spectral–spatial information extraction.

3) With or Without Weighted L_2-Normalization (L_2): By further introducing the L_2-normalization to our multiscale modules, the overall classification accuracies on IP, MUFFL, GFSOFF, Houston2013, and Houston2018 all further increase from 97.95%, 93.04%, 99.19%, 92.38%, and 86.47% to 98.27%, 93.25%, 99.24%, 92.53%, and 86.64%, respectively, as shown in Table XII. Other metrics, i.e., AA and Kappa, generally exhibit the similar improvement when L_2-normalization is introduced. This supports the argument that L_2-normalization helps reduce variation difference between features of different scales by projecting them all

TABLE XII
ABLATION STUDY OF DIFFERENT PARTS OF THE PROPOSED MATA

	MSFE	L_2	WDF	OA (%)	AA (%)	Kappa (%)
IP	×	×	×	95.90	97.41	95.25
	✓	×	×	97.95	99.10	97.62
	✓	✓	×	98.27	99.08	97.99
	✓	×	✓	98.25	99.18	97.97
	✓	✓	✓	98.51	98.19	98.28
MUFFL	×	×	×	91.82	94.70	89.06
	✓	×	×	93.04	95.41	90.69
	✓	✓	×	93.25	95.23	90.96
	✓	×	✓	93.57	95.22	91.39
	✓	✓	✓	93.73	95.24	91.61
GFSOFF	×	×	×	98.72	98.46	98.39
	✓	×	×	99.19	99.43	98.98
	✓	✓	×	99.24	99.47	99.04
	✓	×	✓	99.21	99.29	98.99
	✓	✓	✓	99.27	99.50	99.08
Houston2013	×	×	×	90.97	92.74	90.20
	✓	×	×	92.38	93.73	91.73
	✓	✓	×	92.53	94.12	91.89
	✓	×	✓	93.08	94.59	92.50
	✓	✓	✓	93.83	94.92	93.31
Houston2018	×	×	×	84.68	92.27	80.01
	✓	×	×	86.47	93.54	82.35
	✓	✓	×	86.64	93.55	82.56
	✓	×	✓	87.10	93.66	83.17
	✓	✓	✓	87.88	93.60	84.54

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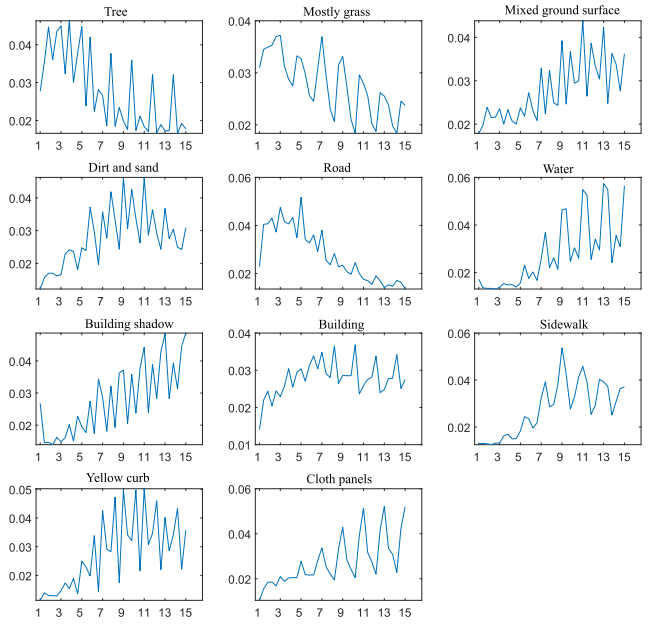


Fig. 15. Classwise decision weights of the proposed MATA derived by training versus multiscale areas on the MUFFL dataset (the areas are represented by their patch size on the horizontal coordinate, and the interval between two patch sizes includes the up, down, left and right parts of the area with the larger patch size).

to sphere of high dimension, and thus helps achieve better classification performance.

4) With or Without WDF: The classwise decision weighting strategy is also important for the proposed model, as listed in Table XII. After introducing it, the overall classification accuracies on the IP, MUFFL, GFSOFF, Houston2013, and Houston2013 datasets are 98.51%, 93.73%, 99.27%, 93.83%, and 87.88%, respectively, which are all better than the situation when average decision strategy is used. In particular, the improvement on the Houston2013 dataset is significant, up to 1%. This indicates that classwise decision weighting strategy is beneficial to process and fuse the semantic information generated by different scales effectively by treating different classes differently since they may have different sizes and are in different areas of HSI patches.

To visually demonstrate the effectiveness of the proposed WDF, classwise weights of the proposed MATA derived by training versus multiscale areas on MUFFL dataset are depicted in Fig. 15. In the horizontal axis, an area with a certain patch size is represented by its patch size, and the up, down, left, and right parts of the area are represented by the values in sequence between its patch size (e.g., x) and a smaller patch size (e.g., $x - 2$). It is clearly shown that different classes have different weights for the decision information from different scales. Many classes give more weights to the semantic information of the larger scales, which indicates that more spatial information is needed to make correct identification of these classes. For example, the weights of the class Building shadow increase with the increase in scales, which means that the more semantic information lies in the large scales. By comparison, some classes provide more weights to the smaller sizes. For example, the weights of the class Road increase from scales 1 to 3 and then decrease with the increase in scales, which demonstrate that less spatial information can help identify road clearly, and more spatial information does not contain more semantic information and even prevent it from correct identification. In addition, some classes give more weights to a medium size. For example, the patch scale corresponding to the largest weight of the class Sidewalk is 9, and the weight increases and decreases with the increase in size, before and after patch scale 9, respectively. It indicates that the most important semantic information lies in the patch scale 9. Overall, different scales have different degrees of semantic information for different classes, and this confirms that our classwise weight decision strategy can help fuse more effective decisions by giving different weights to the decisions from different scales on a per-class basis.

E. Classification Performance in Different Training Ratios

To compare the classification performance of the proposed MATA with other models in different numbers of training samples, Fig. 16 illustrates their OAs on IP, MUFFL, GFSOFF, Houston2013, and Houston2018 with the increase in training ratio. The training ratio represents the ratio of selected samples from all the training samples listed in Tables I–V and is in the set {20%, 40%, 60%, 80%, 100%} for all the five datasets. As shown in Fig. 16, the main trends of OAs of all the models on all the five datasets increase with the number of training samples, and the OAs of the proposed MATA are always higher than the other state-of-the-art models. It achieves above 91% after 80% training ratio on MUFFL and reaches above 95% after 60% training ratio on GFSOFF. In particular, the proposed MATA has higher accuracy on Houston2013 than the other models at various training ratios, and the gap is significant. Overall, the proposed MATA outperforms the other models consistently.

F. Complexity Analysis

To demonstrate the efficiency of MATA, we compare its computational complexity with other models; Table XIII presents the training and testing times for each model on three datasets using PyTorch in Python on a 12th Gen Intel[1] Core[2] i7-12700KF with 32-GB RAM and NVIDIA Geforce RTX 4090 with 24-GB RAM. The single-scale models, CNNHSI and SSCNN, have the lowest training and testing costs due to their small numbers of feature extraction layers. On the other hand, 3DCNN and BERTHSI have higher training costs because of their higher complexity; 3DCNN uses 3-D convolution, which consumes more computation, and BERTHSI's self-attention modules require significant time to calculate the correlation of queries and keys. Among the multiscale models, PMCN has the highest training and testing costs due to its many feature extraction layers that use 3-D convolution. DRCNN follows PMCN, as it contains two stages, i.e., pre-training and fine-tuning. In contrast, the proposed MATA has shorter training and testing times than PMCN and DRCNN,

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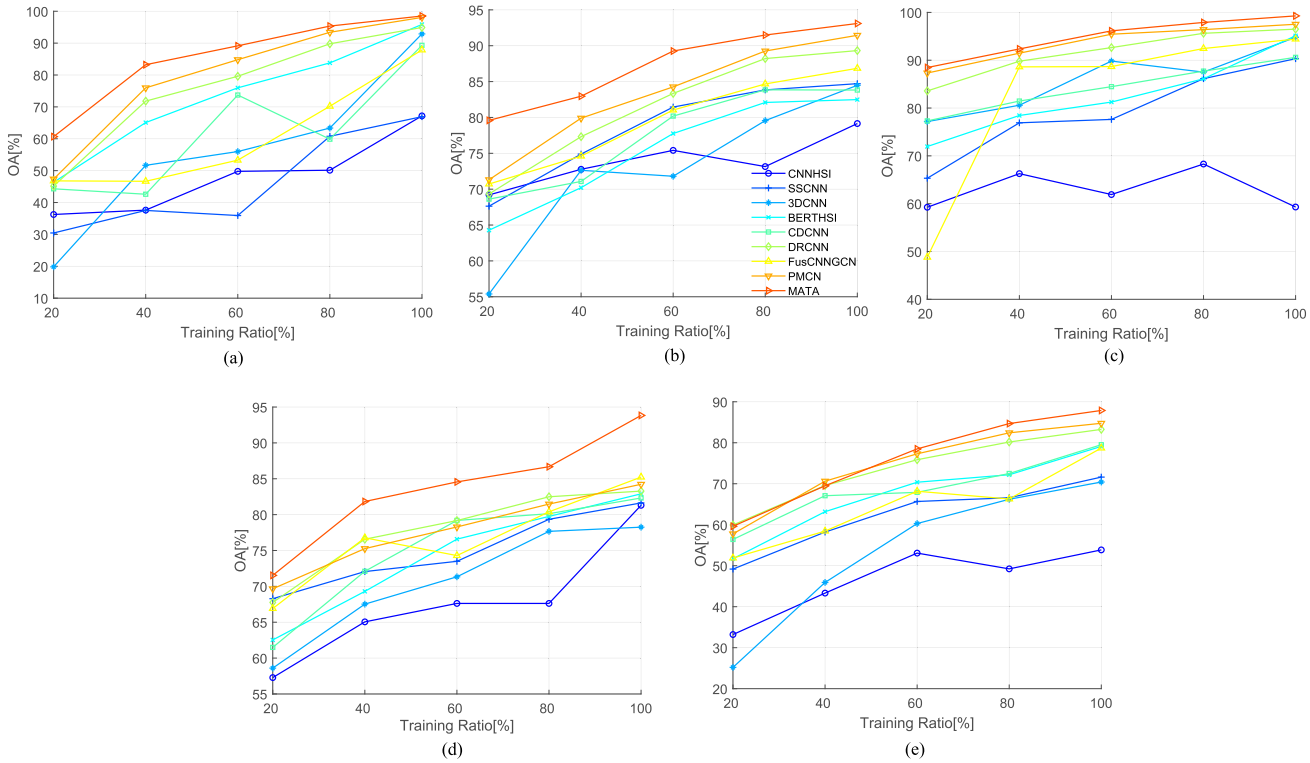


Fig. 16. OA [%] versus training ratio [%] on the HSI datasets. (a) IP. (b) MUFFL. (c) GFSOFF. (d) Houston2013. (e) Houston2018.

TABLE XIII

COMPUTATION TIME [SECOND] AND TOTAL PARAMETERS [k] OF DIFFERENT MODELS FOR IP, MUFFL, GFOSFF, HOUSTON2013, AND HOUSTON2018

Datasets		CNNHSI [18]	SSCNN [21]	3DCNN [22]	BERTHSI [23]	CDCNN [30]	DRCNN [17]	FusCNNGCN [33]	PMCN [34]	MATA
IP	Training time (s)	17.84	15.9	176.01	81.38	16.21	148.54	86.85	1037.51	128.38
	Testing time (s)	0.98	0.92	3.3	3.17	0.89	3.3	16.3	9.2	4.1
	Total parameters (k)	72.18	287.24	1566.57	183.52	1064.37	9157.26	146.11	228.35	884.34
MUFFL	Training time (s)	26.74	20.86	83.41	103.32	17.88	229.43	36.51	1957.92	51.73
	Testing time (s)	1.75	1.84	6.15	6.22	1.83	4.66	7.7	27.72	3.15
	Total parameters (k)	59.37	92.83	451.81	169.41	454.43	6956.99	88.62	142.77	632.87
GFSOFF	Training time (s)	151.96	163.17	1035.31	660.4675	117.48	2077.22	612.23	6655.27	415.60
	Testing time (s)	20.3	17.04	78.38	68.75	19.41	47.91	75.8	149.41	42.70
	Total parameters (k)	75.59	180.33	951.65	175.61	727.84	7901.06	114.25	181.08	727.08
Houston2013	Training time (s)	108.21	44.21	268.79	153.07	33.75	631.61	79.87	4434.06	118.90
	Testing time (s)	30.92	23.51	108.42	69.38	24.11	76.69	83.05	253.44	57.72
	Total parameters (k)	87.96	206.96	1107.69	177.82	813.36	8287.74	122.58	193.17	757.81
Houston2018	Training time (s)	28.86	27.33	114.99	127.21	20.7	329.23	237.14	982.79	239.88
	Testing time (s)	59.65	29.76	96.69	53.79	33.24	82.14	954.67	155.56	72.79
	Total parameters (k)	46.24	70.72	321.9	168.72	383.93	4028.44	83.09	133.29	612.92

demonstrating its superiority in the architecture design of multiscale models that use one shared FE and one shared classifier.

V. CONCLUSION

In this study, an effective and efficient multiscale model, called MATA, was proposed for HSI classification. The MATA model uses a shared FE and classifier architecture to effectively capture multiscale spectral-spatial information while reducing computational complexity. Specifically, MSTAM was incorporated in the FE to extract spectral–spatial information from target pixels and their similar pixels across multiscale areas. To address discrepancies between features of different scales, L_2-normalization was used. For the classifier, a class-wise decision weighting strategy was adopted that accounts for the varying sizes of different classes and the different contributions of semantic features at each scale to each class. The experimental results on three public HSI datasets have demonstrated the superior performance of the proposed MATA compared with existing single- and multiscale models, validating its effectiveness and efficiency in hyperspectral image classification.

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